INCH-POUND

MIL-DTL-25867E
10 July 2019
SUPERSEDING
MIL-DTL-25687D
9 November 2006

DETAIL SPECIFICATION

GAGE, PRESSURE, DIAL INDICATING, PNEUMATIC

Reactivated after 9 November 06 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

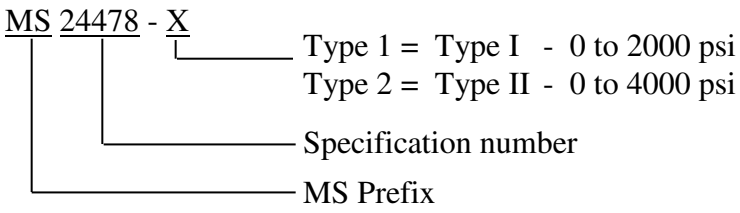
1. SCOPE

1.1 Scope. This specification covers light-weight dial-indicating pneumatic pressure gages, suitable for mounting in pneumatic systems and hydraulic system accumulators of aircraft.

1.2 Classification. The gages are of the following types, as specified (see 6.2):

Type I - 0 to 2000 psi (MS24478-1)
Type II - 0 to 4000 psi (MS24478-2)

1.3 Part or identifying number (PIN). The PIN to be used for pneumatic pressure gages acquired to this specification is created as follows:



Comments, suggestions, or questions on this document should be addressed to DLA Aviation - VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-PRF-6085	- Lubricating Oil: Instrument, Aircraft, Low Volatility.
MIL-A-8625	- Anodic Coatings for Aluminum and Aluminum Alloys.
AND10050	- Bosses, Standard Dimensions for Gasket Seal Straight Thread.
MS28105	- Cover Glass, Aircraft Instrument Dial.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-810	- Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1916	- DoD Preferred Methods for Acceptance of Product.
MS24478	- Gage, Single 1 Inch Dial, Pneumatic Pressure.

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-781	- Reliability Test Methods, Plans, and Environments for Engineering, Development Qualification, and Production.
MIL-HDBK-831	- Preparation of Test Reports.

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(Copies of these documents are available at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

STANDARDIZATION DOCUMENTS

SD-6 - Provisions Governing Qualification.

(Copies of these documents are available at https://quicksearch.dla.mil/.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

ASME INTERNATIONAL

ASME Y14.100 - Engineering Drawing Practices.

(Copies of these documents are available online at http://www.asme.org/.)

SAE INTERNATIONAL

SAE AMS-QQ-P-416 - Plating, Cadmium (Electrodeposited).
SAE AS8879 - Screw Threads - UNJ Profile, Inch Controlled Radius Root with Increased Minor Diameter.
SAE AMS-STD-595 - Colors Used in Government Procurement

(Copies of these documents are available online at http://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The gages furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.6 and 6.4).

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3.2 Materials.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the gage except where magnetic materials are essential.

3.2.3 Nonferrous materials. Nonferrous materials shall be used for all parts of the gage except where ferrous materials are essential.

3.2.4 Metals. Metals shall be corrosion-resistant or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.2.4.1 Dissimilar metals. Unless protected against galvanic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.2.5 Toxic and corrosive fumes. The gage shall be of such material and construction that when the material is burned, fumes or gases that combine with the atmosphere to form acid or corrosive alkali will not be released.

3.2.6 Protective treatment. When materials are used in the construction of the gage that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.2.7 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3 Design and construction. The gage shall be designed to indicate directly the pressure in aircraft hydraulic system accumulators and pneumatic systems. The gage shall be constructed so that no parts will work loose in service and shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.3.1 Interface. Pivots, bearings, and gears shall neither bind nor shake and be as smooth.

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3.3.2 Adjustment mechanism. If the gage is provided with a means for adjusting or correcting the indications, it shall be simple to manipulate and designed so that adjustments can be made without the use of special tools. The means of adjustment need not be outside the case.

3.3.3 Gage mechanism. The gage mechanism shall be comprised of units distinct from the case. The gage shall be sufficiently rugged so that small distortions of the case will not affect accuracy of the indication.

3.3.4 Case. The case dimensions shall be in accordance with MS24478. The case body shall be made of nonferrous low-density metal, shall be uniform in texture, shall have a smooth surface, and shall be manufactured to be rain tight.

3.3.4.1 Vent. The case shall be vented with a hole having a diameter of at least 0.0675 inch. If the hole is plugged, it shall be filled or closed with a permanent porous plug or sealing device that will release or blow out if the pressure sensing element in the gage should fail. There shall be no failure of the cover glass or any portion of the gage casing due to failure of the internal sensing element.

3.3.5 Mounting nuts. When attached insert nuts are used, each mounting lug shall be fitted with 0.138-32 UNC-2B self-locking insert nuts that shall be attached to prevent loosening under an axial load of 25 pounds and to prevent turning under a torque of 10 inch-pounds.

3.3.6 Boss. The boss on the gage shall conform to AND10050. Flats shall be provided on the side of the boss for a minimum distance of 3/8 inch to permit use of a wrench to tighten the mating fittings.

3.3.7 Restrictor. The gage shall be provided with a restrictor orifice having an internal diameter of 0.008 ± 0.0015 inch, a maximum length of 0.156 inch, and a minimum length of 0.0625 inch.

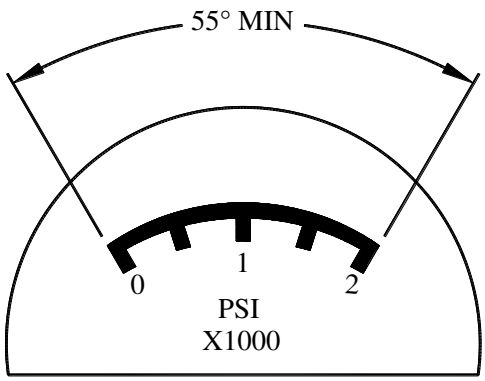
3.3.8 Overpressure stop. The gage shall be provided with an overpressure stop designed to restrain the prime mover thereby limiting the pointer travel to a minimum of 0.062 inch beyond the full scale graduation.

3.3.9 Pointer. A pointer shall be provided and shall have a width at the tip end of 0.017 inch ± 0.003 inch and an increasing taper of approximately 0.031 inch per inch or a width of 0.031 inch ± 0.008 inch along its entire length. The length of the pointer shall be such that the tip will extend into the scale a distance of one-third to two-thirds the length of the shortest graduation.

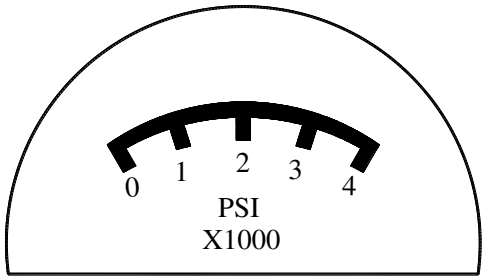
3.3.9.1 Pointer finish. The visible portion of the pointer shall be finished with lusterless white, color identification (ID) 37875 of SAE AMS-STD-595.

3.3.10 Dial. The dial shall be constructed of metal and the dial configuration shall conform to figure 1. The calibration arc for each type specified herein shall be a minimum of 55 degrees on a 0.75 inch radius with the pivot point located on the vertical center line, 0.406 inch below the horizontal centerline.

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2,000 PSI GAGE



4,000 PSI GAGE

FIGURE 1. Dial configuration.

3.3.10.1 Dial markings. The dimensions and colors of the dial markings shall be in accordance with table I. Color identification shall conform to SAE AMS-STD-595.

TABLE I. Dial marking dimensions and colors.

Marking	Height or length, inch ± 1/64	Width of line or graduation, inch ± 0.005	SAE AMS-STD-595 color identification
Numerals 0, 1, 2, 3, 4	5/64	0.016	37875
Major graduations	1/16	0.031	37875
Minor graduations	3/64	0.031	37875
Scale arc	--	0.031	37875
Lettering PSI and X1000	3/64	0.010	37875
Background dial	--	--	37038

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3.3.10.2 Cover glass. The cover glass shall be in accordance with MS28105.

3.3.10.3 Dial-to-cover glass distance. The distance between the inner surface of the cover glass and the dial shall be the minimum practicable and shall not exceed 0.200 inch.

3.3.11 Reliability. The gage shall have a minimum mean-time-between-failures (MTBF) of 2,000 hours based upon a discrimination ratio of 1.5 and a confidence factor of 90 percent when tested as specified in 4.5.17.

3.4 Performance. The gage shall operate satisfactorily after subjection to the following conditions:

a. Operating temperatures ranging from -54 ºC to 100 ºC (-65 ºF to 212 ºF) and non-operating temperatures ranging from -65 ºC to 200 ºC (-85 ºF to 393 ºF).
b. Vibration of 30g from 5 Hz to 5,000 Hz.
c. Temperature-altitude-pressure altitude of 100,000 feet at -55 ºC (-67 ºF).
d. Exposure to salt sea atmosphere for 48 hours without evidence of corrosion or deterioration.
e. Relative humidity of 95 percent.
f. Fungus growth as encountered in tropical climates.
g. Exposure to dry dust-laden atmosphere.
h. Overpressure of 135 percent.

3.4.1 Tolerances. The gage shall operate within the tolerances specified in table II.

TABLE II. Scale error.

Gage type	Applied test pressure PSI	Room temp after all tests except endurance	Tolerance ±PSI		Friction error position error leakage
			High temp (71 ºC) low temp (-54 ºC) and after overpressure and endurance	Low temp exposure (-65 ºC) and temp-alt high temp exposure (200 ºC)	
I	0	75	100	200	--
	500	75	100	175	40
	1,000	75	100	200	--
	1,500	75	100	200	--
	2,000	75	125	225	--
II	0	150	200	400	--
	1,000	150	200	350	80
	2,000	150	200	400	--
	3,000	150	200	400	--
	4,000	150	250	450	--

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3.4.2 Leakage. Leakage shall not exceed the tolerances specified in table II.

3.5 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of ASME Y14.100 shall govern the manufacturer's part numbers and changes of the part numbers.

3.6 Screw threads. Screw threads shall be in accordance with SAE AS8879.

3.7 Lubricants. Lubricants shall conform to MIL-PRF-6085.

3.8 Weight. The weight of the gage shall not exceed 0.18 pound.

3.9 Finishes and protective coatings.

3.9.1 Aluminum-alloy parts. Aluminum-alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

a. Dials, small holes, and case inserts need not be anodized.
b. Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-DTL-5541.
c. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-DTL-5541 may be used in lieu of anodizing.
d. Castings containing non-aluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-DTL-5541 in lieu of anodizing.
e. When abrasion resistance is a factor, chemical films in accordance with MIL-DTL-5541 shall not be used in lieu of anodizing.

3.9.2 Steel parts. Cadmium plating of steel parts, when used, shall be in accordance with SAE AMS-QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10.1 Nameplate. The nameplate shall be located behind the cover glass as shown on MS24478 to provide maximum protection and shall be marked as follows:

Gage, Pressure, Pneumatic
MS type number (as applicable)
Manufacturer's part number (as applicable)
Manufacturer's name or trademark

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3.11 Workmanship. The gage shall be constructed and free of defects such as burrs and sharp edges. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies.

3.11.1 Gears. Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.11.2 Screw assemblies. Assembly screws and bolts shall be tight, meaning the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

3.11.3 Cleaning. The gage shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign materials after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall also be removed.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions.

4.2.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) at room temperature, approximately 25 ºC (77 ºF). When tests are made with atmospheric pressure or room temperature differing from the above values, proper allowance shall be made for the difference from the specified condition.

4.2.2 Attitude. Unless otherwise specified, the gage shall be tested with the mounting surfaces in an upright position.

4.2.3 Tapping. Unless otherwise specified, the gage shall be lightly tapped or vibrated before each reading.

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4.3 Qualification inspection. The qualification inspection shall consist of all the tests specified in 4.5.

4.3.1 Test samples. The test samples shall consist of 13 gages representative of the production equipment. The samples shall be identified with the manufacturer's part number and any additional information required by the contracting official. Ten of the samples shall be subjected to only the reliability test specified in 4.5.17. Three shall be subjected to all the tests except reliability. One of these shall be subjected to the vent test specified in 4.5.18.

4.3.2 Test report. The contractor shall prepare and submit a test report. MIL-HDBK-831 may be used for guidance.

4.4 Conformance inspection. The conformance inspection shall consist of individual tests and sampling plan A or B inspections when specified in the contract.

4.4.1 Individual tests. Each production gage shall be subjected to the following tests as described under 4.5:

a. Examination of product (see 4.5.1).
b. Scale and friction error (see 4.5.2).
c. Position error (see 4.5.3).

4.4.2 Sampling plan inspections.

4.4.2.1 Sampling plan A. Sampling and inspection shall be in accordance with MIL-STD-1916 for verification level I. Sampling plan A inspection shall consist of the following tests:

a. Individual tests (see 4.4.1).
b. Vibration (see 4.5.4).
c. Low temperature (see 4.5.5).
d. High temperature (see 4.5.6).

4.4.2.2 Sampling plan B. Unless otherwise specified, three gages shall be selected at random from the first 15 gages produced on the contract or acquisition order and subjected to the following tests:

a. Sampling plan A tests (see 4.4.2.1).
b. Seasoning (see 4.5.7).
c. Endurance (see 4.5.8).
d. Low pressure (altitude) (see 4.5.9).
e. Low temperature exposure (see 4.5.10).
f. High temperature exposure (see 4.5.11).
g. Humidity (see 4.5.12).
h. Salt-fog (see 4.5.13).
i. Fungus (see 4.5.14).
j Sand and dust (see 4.5.15).
k. Overpressure (see 4.5.16).

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4.5 Test methods.

4.5.1 Examination of product. Each gage shall be inspected to determine compliance with the requirements specified herein with respect to materials, markings, dimensions, weight, axial load, torque, and workmanship.

4.5.2 Scale and friction error. The gage shall be tested for scale error at ambient temperature at the test points specified in table II. The gage shall be lightly tapped or vibrated before each reading to remove normal friction. The difference between the readings before and after the gage is tapped or vibrated shall constitute friction error. The scale and friction errors shall not exceed the tolerances specified in table II. This test shall be conducted both up-scale and down-scale at the specified test points.

4.5.3 Position error. The gage shall be pressurized to obtain a scale reading of approximately one-half scale. The maximum change in indication produced by rotating the gage from the normal operating position to a position 90 degrees both clockwise and counterclockwise (longitudinal axis vertical) shall not exceed the position error tolerance specified in table II.

4.5.4 Vibration. The gage shall be subjected to a vibration test in accordance with Method 514, Procedure I, Equipment categories 13 and 22 of MIL-STD-810. An indication of approximately mid-scale shall be maintained on the gage during this test. At the conclusion of this test, the gage shall be subjected to the scale and friction error test. The errors shall not exceed the tolerances specified in table II. No damage to the gage shall result from this test.

4.5.5 Low temperature. The gage shall be subjected to a low temperature test in accordance with Method 502, Procedure I of MIL-STD-810 at -54 ºC ± 2 ºC (-65 ºF ± 4 ºF) for a period of 4 hours. During this test, an indication of approximately mid-scale shall be maintained on the gage. At the end of this period and while still at the specified temperature, the gage shall be subjected to the scale and friction error test. The errors shall not exceed the tolerances specified in table II.

4.5.6 High temperature. The gage shall be subjected to a high temperature test in accordance with Method 501, Procedure I of MIL-STD-810 at 100 ºC ± 2 ºC (212 ºF ± 4 ºF) for a period of 6 hours. During this test an indication of approximately mid-scale shall be maintained on the gage. At the end of the 6-hour period and while still at the specified temperature, the gage shall be subjected to the scale and friction error test. The tolerances shall not exceed those specified in table II.

4.5.7 Seasoning. The gage shall be operated for 10,000 cycles at a rate of 30 cycles ± 10 cycles per minute (cpm) from zero indication to 80 percent ± 5 percent of full scale and back to zero indication. Within one hour after the conclusion of this test, the gage shall be subjected to the scale error test. The scale errors shall not exceed the tolerances specified in table II. The gage shall then be examined to determine that no undue wear has occurred.

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4.5.8 Endurance. The gage shall be subjected to an endurance test in accordance with 4.5.7 except that it shall be operated for 60,000 cycles in lieu of 10,000 cycles.

4.5.9 Low-pressure (altitude). The low-pressure (altitude) test shall be conducted in accordance with Method 500, Procedure I of MIL-STD-810, except the temperature shall be -55 ºC (-67 ºF) with the chamber pressure equivalent to an altitude of 100,000 feet. The rate of pressure change shall not exceed an average of 3 feet per second while under these conditions. At the end of a 1-hour period, a scale and friction error test shall be conducted. The errors shall not exceed the tolerances specified in table II. This test may be conducted concurrently with the low temperature exposure test.

4.5.10 Low temperature exposure. The low temperature exposure test shall be conducted in accordance with Method 502, Procedure I of MIL-STD-810. The gage shall be exposed to a temperature of -65 ºC ± 2 ºC (-85 ºF ± 4 ºF) for a minimum of 48 hours after which it shall be operated throughout its range while at the specified temperature. The temperature shall then be raised to -54 ºC ± 2 ºC (-65 ºF ± 4 ºF) and the gage maintained at this temperature for 24 hours. While at -54 ºC ± 2 ºC (-65 ºF ± 4 ºF), the gage shall be subjected to the scale error at room temperature and friction tests. The tolerances shall be within those specified in table II. The gage shall be returned to room temperature for a minimum period of 4 hours and then subjected to the scale error test. The scale errors shall not exceed the tolerances specified in table II for room temperature.

4.5.11 High temperature exposure. The gage shall be placed in a temperature chamber and subjected to 200 ºC ± 2 ºC (392 ºF ± 4 ºF) for a period of 24 hours. It shall then be returned to room temperature for a minimum of 4 hours after which it shall be subjected to the scale error test at room temperature. The scale error shall not exceed the room temperature tolerances specified in table II.

4.5.12 Humidity. The gage shall be subjected to a humidity test in accordance with Method 507, Procedure I of MIL-STD-810. External connections to the gage shall be made to simulate installed conditions. Immediately after completion of this test, the gage shall be subjected to the scale and friction error test. The scale and friction error shall not exceed the tolerances specified in table II.

4.5.13 Salt-fog. The gage shall be subjected to a salt-fog test for a period of 48 hours in accordance with Method 509, Procedure I of MIL-STD-810. There shall be no evidence of external corrosion or deterioration that might affect subsequent operation. The scale errors after this test shall not exceed the tolerances specified in table II for room temperature scale error.

4.5.14 Fungus. The gage shall be subjected to a fungus test in accordance with Method 508, Procedure I of MIL-STD-810 after completion of qualification testing and after the gage has been subjected to the salt-fog test. The gage shall be disassembled so that the internal mechanism will be exposed to the fungus spores. At the end of the test period the gage shall be examined to determine that no fungus growth has occurred.

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4.5.15 Sand and dust. The gage shall be subjected to a sand and dust test in accordance with Method 510, Procedure I of MIL-STD-810. At the completion of this test the gage shall be removed from the test chamber, allowed to cool to room temperature, and then subjected to the scale error test (at room temperature) and friction error test. The scale error and friction error shall not exceed the tolerances specified in table II.

4.5.16 Overpressure. The gage shall be subjected to an overpressure of 135 percent of the rated pressure for a period of 10 minutes. Not less than 1 hour after application of the specified pressure, the gage shall be subjected to the scale error test at room temperature. The scale error shall not exceed the tolerance specified in table II.

4.5.17 Reliability. Ten gages shall be tested for reliability. Recording, data handling, and reporting shall be performed for the government. MIL-HDBK-781 may be used for guidance.

4.5.17.1 Reliability test. The test equipment and fixtures shall include provisions for vibration and temperature cycling at periodic intervals specified herein. During this test, input pressure shall be applied to the gage at a rate of 2 cpm to 3 cpm from 0 percent to 85 percent of full scale and return. Completion of 4 hours of operation shall include at least 480 operational cycles. For at least 10 minutes out of every hour of "ON" time, the gage shall be vibrated at 2.2g ± 10 percent peak acceleration value at any non-resonant frequency between 20 cycles per second (cps) and 60 cps, measured at the mounting points on the equipment. The temperature cycles shall be conducted in the following sequence: 4 hours at -55 ºC (-67 ºF), temperature change to 50 ºC (122 ºF) at a minimum rate of 5 ºC (41 ºF) per minute; 4 hours at 50 ºC (122 ºF), temperature change to -55 ºC (-67 ºF), and 4 hours at -55 ºC (-67 ºF). In case of malfunction, the malfunction shall be recorded and the test interrupted while repairs are made. Upon completion of each 250 hours of cycling, the gage shall be checked for scale error. The scale error reading shall not exceed twice the room temperature tolerance in table II. No calibration shall be performed on the units during this test. The test may be interrupted for weekend shutdown. The contractor shall prepare a report that contains test procedures, details of any malfunctions, and corrective actions taken. MIL-HDBK-781 may be used for guidance.

4.5.18 Vent. If a plug is used, one gage shall be subjected to a pressure sufficient to blow out the plug or sealing device that fills the vent hole. The pressure shall be applied to the inside of the case after the sensing element has been ruptured mechanically. The cover glass and case shall not be ruptured or damaged as a result of this test.

4.6 Maintenance of qualification. At specified intervals determined by the qualifying activity, the manufacturer must be able to demonstrate that the company still has the capabilities and facilities necessary to produce the QPL items in accordance with this specification and in accordance with the provisions governing qualification specified in SD-6.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The gages covered by this specification are intended for use in indicating directly the pressure in hydraulic system accumulators in aircraft pneumatic systems and in locations having ambient temperatures ranging from -54 ºC to 100 ºC (-65 ºF to 212 ºF).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of gage desired (see 1.2).
c. The specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
d. Packaging required (see 5.1).

6.3 Safety Data Sheet (SDS). If the use of cadmium is not considered an article according to OSHA's guidance for hazard determination, a SDS must be prepared and submitted in accordance with FED-STD-313 (obtain from https://assist.dla.mil). The SDS must also meet the requirements of 29 CFR 1910.1200. The 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must be readily available to personnel using the material.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products that are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-25867 whether or not such products have actually been listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the federal government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Aviation -VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or STDZNMGT@dla.mil via email.

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6.5 Subject term (key word) listing.

accumulators
aircraft indicators
cadmium
hydraulic system
pointer
restrictor

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AV
Navy - AS
Air Force - 71

Review Activities:
Army - MI

Preparing Activity:
DLA - GS1

(Project 6685-2019-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at http://assist.dla.mil/.

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